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| EDUCATION               | <b>Ph.D. in Electrical Engineering</b> (minor in CS), Iowa State University, Ames IA, 4.0/4.0<br><b>B. S. in Automation</b> , Zhejiang University, Hangzhou China                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 12/2014<br>06/2009                                                                                                                                                                                                                                                                 |
| RESEARCH AREAS          | <b>Optimal Control and Intelligent Systems</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                    |
| EMPLOYMENT              | <b>Oakland University</b> , Rochester, MI, USA<br>- <b>SECS Distinguished Associate Professor</b><br>- <b>Associate Professor</b> , Department of Electrical and Computer Engineering<br>- Assistant Professor, Department of Electrical and Computer Engineering<br><b>General Motors</b> , Michigan, USA<br>- Senior Control Systems Engineer, Vehicle and Propulsion Control, Milford<br>- Summer Intern, Software V&V, Warren (GM R&D)<br><b>Idaho National Laboratory</b> , Idaho Falls, ID, USA<br>- R&D Scientist, Power and Energy Systems                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 04/2026–present<br>08/2024–present<br>08/2020–08/2024<br><br>01/2017–08/2020<br>04/2014–07/2014<br><br>11/2014–12/2016                                                                                                                                                             |
| HONORS AND RECOGNITIONS | <b>NSF CAREER Award</b> , National Science Foundation<br><b>Best Paper Award</b> , IEEE Transactions on Automation Science and Engineering<br><b>Best Paper Award</b> , IEEE Cyber Awareness & Research Symposium<br><b>Best Paper Award</b> , IEEE International Conference on Electro-Information Technology<br><b>IEEE Senior Member</b><br><b>Associate Editor</b> , IEEE Robotics and Automation Letters<br><b>Associate Editor</b> , IEEE Systems, Man, and Cybernetics Letters<br><b>Associate Editor</b> , IEEE Conference on Decision and Control<br><b>Associate Editor</b> , ASME Journal of Dynamic Systems, Measurement, and Control<br><b>Associate Editor</b> , American Control Conference<br><b>SECS Distinguished Associate Professor Award</b> , Oakland University<br><b>ATS Best Paper Finalist Award</b> , American Control Conference<br><b>Outstanding Graduate Mentor Award</b> , Oakland University<br><b>Most Research Active Award</b> , Oakland University<br><b>New Investigator Research Excellence Award</b> , Oakland University<br><b>Most Active Grant Seeker Award</b> , Oakland University<br><b>Oakland County 40 Under 40</b> , Oakland County<br><b>NSF CMMI Game Changer Academies (C-GCA) Panel Fellow</b> , NSF<br><b>Faculty Recognition Award for Research</b> , Oakland University<br><b>R&amp;D 100 Award</b> , R&D World<br><b>Associate Editor</b> , IEEE International Conference on Robotics and Automation<br><b>INL Publication Achievement Award</b> , Idaho National Laboratory<br><b>INL Exceptional Contributions Program Award</b> , Idaho National Laboratory<br><b>Research Excellence Award</b> , Iowa State University<br><b>Student Travel Award</b> , American Control Conference<br><b>Third Class Scholarship for Undergraduate Student</b> , Zhejiang University<br><b>Outstanding Student</b> , Zhejiang University | 2022<br>2016<br>2025<br>2023<br>2020<br>2026–present<br>2026–present<br>2026–present<br>2025–present<br>2024–present<br>2026<br>2026<br>2025<br>2024<br>2024<br>2024<br>2023 & 2024<br>2024<br>2024<br>2023<br>2023<br>2020<br>2016<br>2015 & 2016<br>2014<br>2014<br>2008<br>2008 |
| HONORS BY MY ADVISEES   | <b>ATS Best Paper Finalist Award</b> (Luke Nuculaj), American Control Conference<br><b>Thesis with Distinction</b> (Briana Popa), Oakland University Honor College                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 2026<br>2026                                                                                                                                                                                                                                                                       |

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|--------------------------------------------------------------------------------------|------|
| <b>YHC Thesis Grant</b> (Briana Popa), Oakland University Honor College              | 2025 |
| <b>Student Grant</b> (Wanqun Yang), Modeling, Estimation and Control Conference      | 2025 |
| <b>SECS Best Graduate Paper Award</b> (Zhaodong Zhou), Oakland University            | 2024 |
| <b>Best Paper Award</b> (Zhaodong Zhou), IEEE EIT Conference                         | 2023 |
| <b>Undergraduate Research Competition Winner</b> (Steven DeCoste), ASME ICE Division | 2022 |

#### SELECTED GRANTS

Total: ~\$5.8m; As lead/sole PI: ~\$2.2m; Personal share: ~\$2.4m

- [9] **Jun Chen**, Dan Aloï and Zhijun Jason Wu, “Oakland University Team for SAE CivicProgress Challenge,” 09/2026-08/2028, General Motors and SAE International.
- [8] **Jun Chen**, “I-Corps: Sensor-lean Battery Management Systems,” 04/2026–03/2027, NSF-TIP.
- [7] **Jun Chen**, “Collaborative Research: Scalable Privacy Verification and Quantification for Multi-Robot Systems,” 10/2025–09/2028, NSF-FRR. (Collaborative project with Dr. Feng Lin from Wayne State University; Lead institution – OU)
- [6] **Jun Chen**, “Sensor-lean Estimation and Monitoring for Second Life EV Batteries,” 05/2025–04/2028, NSF-ECCS-EPCN.
- [5] **Jun Chen**, “CAREER: Reconfigurable and Predictive Control with Reinforcement Learning Supervisor for Active Battery Cell Balancing,” 01/2023–12/2027, NSF-ECCS-EPCN.
- [4] Ankun Yang, **Jun Chen**, Xia Wang, Yongsoon Yoon, Dan DeVescovo, Daniel Aloï, Shadi Alawneh and Mohammad Sedigh Toulabi, “Next-Gen Electrification Testing and Standards Facility: from Materials to Vehicles,” 10/2024–09/2026, National Institute of Standards and Technology.
- [3] **Jun Chen**, “Sensor Reduction for Large Battery Packs,” 09/2024–08/2026, Michigan Translational Research and Commercialization (MTRAC) for Advanced Transportation Innovation Hub.
- [2] **Jun Chen**, “Affordable Autonomous Vehicle Platform for University Research and Education,” 09/2025–08/2026, MEDC ADVANCE Proof-of-Concept Fund.
- [1] Yang Chen, **Jun Chen** and Om Prakash Yadav, “Ocean Energy Supported Multi-Energy System Planning and Operation Optimization for Sustainable Coastal Community,” 07/2023–06/2024, Coastal Studies Institute, North Carolina Renewable Ocean Energy Program.

#### PUBLICATIONS

(h-index: 28; underline: students under my close supervision; \*: corresponding author)

#### Patents

- [4] Min Sun, Yiran Hu, David Edwards, **Jun Chen**, Insu Chang and Steven Moorman, “[Active Thermal Management System and Method for Flow Control](#),” U.S. Patent No. US11312208 B2, April 26, 2022.
- [3] **Jun Chen**, Ruixing Long and Yiran Hu, “[Method for Increasing Control Performance of Model Predictive Control Cost Functions](#),” U.S. Patent No. US11192561 B2, December 7, 2021.
- [2] Yiran Hu, David Edwards, Michael Paratore Jr, Min Sun, **Jun Chen**, Eugene Gonze and Sergio Quelhas, “[Method and Apparatus for Control of Propulsion System Warmup Based on Engine Wall Temperature](#),” U.S. Patent No. 11078825 B2, August 3, 2021.
- [1] **Jun Chen**, David Edwards, Yiran Hu, Min Sun, Adam J. Heinzen and Michael A. Smith, “[Method and System for Determining Thermal State](#),” U.S. Patent No. 10995688 B2, May 4, 2021.

#### Book Chapters

- [1] Mariam Ibrahim, **Jun Chen** and Ratnesh Kumar, “[Quantification of Centralized/Distributed Secrecy in Stochastic Discrete Event Systems](#),” in *Recent Advances in Systems Safety and Security*, Editors: Emil Pricop and Grigore Stamatescu, Springer, May 2016, ISBN: 978-3-319-32523-1.

#### Journal Articles

- [56] Guojiang Xiong\*, Sha Yang, **Jun Chen** and Ali Wagdy Mohamed, “[Reinforcement Learning Whale Optimization Algorithm-Based Optimal Equivalent Circuit Models for Photovoltaic Cells and Modules](#),” *Journal of Big Data*, (Accepted for Publication May 2026).

- [55] [Wanqun Yang](#), [Ranya Badawi](#) and **Jun Chen\***, “Event-Triggered Model Predictive Control of a Buck Converter with Kalman Filter State Estimation,” *Journal of Dynamic Systems, Measurement, and Control*, (Accepted for Publication April 2026).
- [54] [Ali Irshayyid](#), [Wanqun Yang](#) and **Jun Chen\***, “Real-time Balancing Control of Reconfigurable Battery Packs using Reinforcement Learning,” *IEEE Transactions on Transportation Electrification*, (Accepted for Publication March 2026).
- [53] [Ali Irshayyid](#) and **Jun Chen\***, “Surrogate Model for Reconfigurable Battery Packs using Graph Neural Networks,” *IEEE Transactions on Industrial Informatics*, (Accepted for Publication March 2026)
- [52] [Zhaodong Zhou](#) and **Jun Chen\***, “Event-Triggered MPC with Linear Inter-Event Control for AV Path Tracking,” *IEEE Robotics and Automation Letters*, volume 11, number 7, pages 7868–7875, July 2026.
- [51] [Neha Bhatt](#), [Zhaodong Zhou](#), [Krunal Mehta](#) and **Jun Chen\***, “Adaptive Downsampling and Refinement for Fast Iterative Closest Point,” *IEEE Systems, Man, and Cybernetics Letters*, volume 1, number 1, pages 20–26, June 2026.
- [50] [Fei Wang\\*](#), [Feng Lin](#) and **Jun Chen**, “Information Control in Networked Multi-User Discrete Event Systems Using State Estimates,” *IEEE Transactions on Automation Science and Engineering*, volume 23, number 1, pages 11044–11056, June 2026.
- [49] [Hussein Alawsi](#), [Zhaodong Zhou](#), [Ali Irshayyid](#) and **Jun Chen\***, “Automated Parking Systems Using Reinforcement Learning Assisted Model Predictive Control,” *IET Cyber-Systems and Robotics*, volume 8, number 1, pages 1–17, May 2026.
- [48] [Zhaodong Zhou](#) and **Jun Chen\***, “Privacy-Preserving Personalized Autonomous Vehicle Lane Change Using Inverse Reinforcement Learning,” *IEEE Transactions on Vehicular Technology*, volume 75, number 4, pages 5376–5387, April 2026.
- [47] [Nana Duah](#), [Yang Chen\\*](#), [Om Prakash Yadav](#) and **Jun Chen**, “Marine Energy Supported Multi-Energy System Planning and Operation Optimization for Sustainable Coastal Community,” *Sustainable Energy, Grids and Networks*, volume 44, number 1, pages 1–15, December 2025.
- [46] [Luke Nuculaj](#) and **Jun Chen\***, “Simultaneous Cell State Estimation via Dense Adaptive Extended Kalman Filter,” *IEEE Transactions on Control Systems Technology*, volume 33, number 6, pages 2092–2107, November 2025.
- [45] **Jun Chen\*** and [Feng Lin](#), “A General Framework for Detectability in Stochastic Discrete Event Systems,” *IEEE Control Systems Letters*, volume 9, number 1, pages 2465–2470, October 2025.
- [44] [Owais Ogdeh](#), [Luke Nuculaj](#), [Ali Irshayyid](#), [Zhaodong Zhou](#) and **Jun Chen\***, “Cell State-of-Charge Estimation with Limited Voltage Sensor Measurements,” *Applied Sciences*, volume 25, number 18, pages 1–15, September 2025.
- [43] [Wanqun Yang](#) and **Jun Chen\***, “Comparison of Linear MPC and Explicit MPC for Battery Cell Balancing Control,” *Algorithms*, volume 18, number 9, pages 1–13, September 2025.
- [42] [David Flessner](#) and **Jun Chen\***, “Reinforcement Learning-Based Event-Triggered Model Predictive Control for Electric Vehicle Active Battery Cell Balancing,” *ASME Letters in Dynamic Systems and Control*, volume 5, number 2, pages 1–5, April 2025.
- [41] [Zhaodong Zhou](#), [Mingyuan Tao](#), [Jiayi Qiu](#), [Peng Zhang](#), [Meng Xu](#) and **Jun Chen\***, “Autonomous Vehicle Path Tracking Using Event-Triggered MPC with Switching Model: Methodology and Real-World Validation,” *IET Control Theory & Applications*, volume 19, number 1, pages 1–10, January 2025.
- [40] [Ranya Badawi](#) and **Jun Chen\***, “Event-Triggered Boost Converter Model Predictive Control with Kalman Filter,” *Systems Science & Control Engineering*, volume 12, number 1, pages 1–15, December 2024.
- [39] [Guojiang Xiong\\*](#), [Jing Zhang](#), [Xiaofan Fu](#), **Jun Chen** and [Ali Wagdy Mohamed](#), “Seasonal Short-term Photovoltaic Power Prediction Based on GSK-BiGRU-XGboost Considering Correlation of Meteorological Factors,” *Journal of Big Data*, volume 11, number 164, pages 1–19, November 2024.
- [38] [Mingqiang Wang](#), [Lei Zhang\\*](#), **Jun Chen**, [Zhiqiang Zhang](#), [Zhenpo Wang](#) and [Dongpu Cao](#), “A Hybrid Trajectory Prediction Framework for Automated Vehicles with Attention Mechanisms,” *IEEE Transactions on Transportation Electrification*, volume 10, number 3, pages 6178–6194, September 2024.

- [37] [Ali Irshayyid](#), **Jun Chen\*** and Guojiang Xiong, “[A Review on Reinforcement Learning-based Highway Autonomous Vehicle Control](#),” *Green Energy and Intelligent Transportation*, volume 3, number 4, pages 1–19, August 2024.
- [36] Cong Wang, Zhenpo Wang, Lei Zhang\*, **Jun Chen** and Dongpu Cao, “[Post-Impact Stability Control for Road Vehicles: State-of-the-Art Methodologies and Perspectives](#),” *IEEE Transactions on Intelligent Transportation Systems*, volume 25, number 8, pages 8295–8312, August 2024.
- [35] **Jun Chen\***, Aman Behal, Zhaojian Li and Chong Li, “[Active Battery Cell Balancing by Real Time Model Predictive Control for Extending Electric Vehicle Driving Range](#),” *IEEE Transactions on Automation Science and Engineering*, volume 21, number 3, pages 4003–4015, July 2024.
- [34] Zhaodong Zhou, Christopher Rother and **Jun Chen\***, “[Comparison of Two-Wheel and Four-Wheel Steering using Event-Triggered Predictive Motion Control and Scale Vehicles](#),” *ASME Letters in Dynamic Systems and Control*, volume 4, number 3, pages 1–6, July 2024.
- [33] **Jun Chen\***, Lei Zhang and Weinan Gao, “[Reconfigurable Model Predictive Control for Large Scale Distributed Systems](#),” *IEEE Systems Journal*, volume 18, number 2, pages 965–976, June 2024.
- [32] Kaixiang Zhang, Kaian Chen, Zhaojian Li\*, **Jun Chen** and Yang Zheng, “[Privacy-Preserving Data-Enabled Predictive Leading Cruise Control in Mixed Traffic](#),” *IEEE Transactions on Intelligent Transportation Systems*, volume 25, number 5, pages 3467–3482, May 2024.
- [31] David Flessner, **Jun Chen\*** and Guojiang Xiong, “[Reinforcement Learning-based Event-Triggered Active Battery Cell Balancing Control for Electric Vehicle Range Extension](#),” *Electronics*, volume 13, number 5, pages 1–22, March 2024.
- [30] Fengying Dang, Dong Chen, **Jun Chen\*** and Zhaojian Li\*, “[Event-Triggered Model Predictive Control with Deep Reinforcement Learning for Autonomous Driving](#),” *IEEE Transactions on Intelligent Vehicles*, volume 9, number 1, pages 459–468, January 2024.
- [29] Lei Zhang\*, Qi Wang, **Jun Chen**, Zhenpo Wang and Shaohua Li, “[Brake-by-Wire System for Passenger Cars: A Review of Structure, Control, Key Technologies, and Application in X-by-Wire Chassis](#),” *eTransportation*, volume 18, number 1, pages 1–15, October 2023.
- [28] Mohammad R. Hajidavalloo, **Jun Chen\***, Qiuha Hu, Ziyong Song, Xunyu Yin and Zhaojian Li, “[NMPC-based Integrated Thermal Management of Battery and Cabin for Electric Vehicles in Cold Weather Conditions](#),” *IEEE Transactions on Intelligent Vehicles*, volume 8, number 9, pages 4208–4222, September 2023.
- [27] Christopher Rother, Zhaodong Zhou and **Jun Chen\***, “[Development of a Four-Wheel Steering Scale Vehicle for Research and Education on Autonomous Vehicle Motion Control](#),” *IEEE Robotics and Automation Letters*, volume 8, number 8, pages 5015–5022, August 2023.
- [26] Zhaodong Zhou, Christopher Rother and **Jun Chen\***, “[Event-Triggered Model Predictive Control for Autonomous Vehicle Path Tracking: Validation using CARLA Simulator](#),” *IEEE Transactions on Intelligent Vehicles*, volume 8, number 6, pages 3547–3555, June 2023.
- [25] **Jun Chen\***, “[A Probabilistic Test for A-Diagnosability of Stochastic Discrete-Event Systems with Guaranteed Error Bound](#),” *IEEE Control Systems Letters*, volume 7, number 1, pages 2833–2838, June 2023.
- [24] Zaiyu Gu, Guojiang Xiong\*, Xiaofan Fu, Ali Wagdy Mohamed, Mohammed Azmi Al-Betar, Hao Chen and **Jun Chen**, “[Extracting Accurate Parameters of Photovoltaic Cell Models via Elite Learning Adaptive Differential Evolution](#),” *Energy Conversion and Management*, volume 285, pages 1–25, June 2023.
- [23] Qinghua Liu, Guojiang Xiong\*, Xiaofan Fu, Ali Wagdy Mohamed, Jing Zhang, Mohammed Azmi Al-Betar, Hao Chen, **Jun Chen** and Sheng Xu, “[Hybridizing Gaining-Sharing Knowledge and Differential Evolution for Large-scale Power System Economic Dispatch Problems](#),” *Journal of Computational Design and Engineering*, volume 10, number 2, pages 615–631, April 2023.
- [22] [Ali Irshayyid](#) and **Jun Chen\***, “[Comparative Study of Cooperative Platoon Merging Control Based on Reinforcement Learning](#),” *Sensors*, volume 23, number 2-990, pages 1–23, January 2023.
- [21] **Jun Chen\***, Zhaodong Zhou, Ziwei Zhou, Xia Wang and Boryann Liaw, “[Impact of Battery Cell Imbalance on Electric Vehicle Range](#),” *Green Energy and Intelligent Transportation*, volume 1, number 3, pages 1–8, December 2022.

- [20] **Jun Chen\*** and Ratnesh Kumar, “[Stochastic Failure Prognosis of Discrete Event Systems](#),” *IEEE Transactions on Automatic Control*, volume 67, number 10, pages 5487–5492, October 2022.
- [19] **Jun Chen\*** and Junhui Zhao, “[Generating Synthetic Wind Speed Scenarios using Artificial Neural Networks for Probabilistic Analysis of Hybrid Energy Systems](#),” *International Journal of Modelling, Identification and Control*, volume 41, number 3, pages 183–192, July 2022.
- [18] Xuan Xie, Guojiang Xiong\*, **Jun Chen** and Jing Zhang, “[Universal Transparent Artificial Neural Network-Based Fault Section Diagnosis Models for Power Systems](#),” *Advanced Theory and Simulations*, volume 5, number 4, pages 1–12, April 2022.
- [17] Guojiang Xiong\*, Xufeng Yuan, Ali Wagdy Mohamed, **Jun Chen** and Jing Zhang, “[Improved Binary Gaining-sharing Knowledge based Algorithm with Mutation for Fault Section Location in Distribution Networks](#),” *Journal of Computational Design and Engineering*, volume 9, number 2, pages 393–405, April 2022.
- [16] **Jun Chen\*** and Ramesh S, “[Model-based Validation of Diagnostic Software with Application in Automotive Systems](#),” *IET Cyber-Systems and Robotics*, volume 3, number 2, pages 140–149, June 2021.
- [15] **Jun Chen\***, “[Extended Kalman Filter Steady Gain Scheduling using  \$k\$ -means Clustering](#),” *International Journal of Modelling, Identification and Control*, volume 34, number 2, pages 158–162, February 2020.
- [14] Xiang Yin, **Jun Chen**, Zhaojian Li and Shaoyuan Li, “[Robust Fault Diagnosis of Stochastic Discrete Event Systems](#),” *IEEE Transactions on Automatic Control*, volume 64, number 10, pages 4237–4244, October 2019.
- [13] **Jun Chen**, Christoforos Keroglou, Christoforos N. Hadjicostis and Ratnesh Kumar, “[Revised Test for Stochastic Diagnosability of Discrete-Event Systems](#),” *IEEE Transactions on Automation Science and Engineering*, volume 15, number 1, pages 404–408, January 2018.
- [12] **Jun Chen**, Peter Molnar and Aman Behal, “[Identification of a Stochastic Resonate-and-Fire Neuronal Model via Nonlinear Least Squares and Maximum Likelihood Estimation](#),” *International Journal of Modelling, Identification and Control*, volume 28, number 3, pages 221–231, October 2017.
- [11] **Jun Chen** and Cristian Rabiti, “[Synthetic Wind Speed Scenarios Generation for Probabilistic Analysis of Hybrid Energy Systems](#),” *Energy*, volume 120, pages 507–517, February 2017.
- [10] **Jun Chen**, Mariam Ibrahim and Ratnesh Kumar, “[Quantification of Secrecy in Partially Observed Stochastic Discrete Event Systems](#),” *IEEE Transactions on Automation Science and Engineering*, volume 14, number 1, pages 185–195, January 2017.
- [9] Jong S. Kim, **Jun Chen** and Humberto E. Garcia, “[Modeling, Control, and Dynamic Performance Analysis of a Reverse Osmosis Desalination Plant Integrated within Hybrid Energy Systems](#),” *Energy*, volume 112, pages 52–66, October 2016.
- [8] **Jun Chen** and Humberto E. Garcia, “[Economic Optimization of Operations for Hybrid Energy Systems under Variable Markets](#),” *Applied Energy*, volume 177, pages 11–24, September 2016.
- [7] **Jun Chen**, Humberto E. Garcia, Jong S. Kim and Shannon M. Bragg-Sitton, “[Operations Optimization of Nuclear Hybrid Energy Systems](#),” *Nuclear Technology*, volume 195, number 2, pages 143–156, August 2016.
- [6] Humberto E. Garcia, **Jun Chen**, Jong S. Kim, Richard B. Vilim, William R. Binder, Shannon M. Bragg-Sitton, Richard D. Boardman, Michael G. McKellar and Christiaan J. J. Paredis, “[Dynamic Performance Analysis of Two Regional Nuclear Hybrid Energy Systems](#),” *Energy*, volume 107, pages 234–258, July 2016.
- [5] **Jun Chen** and Ratnesh Kumar, “[Fault Detection of Discrete-Time Stochastic Systems Subject to Temporal Logic Correctness Requirements](#),” *IEEE Transactions on Automation Science and Engineering*, volume 12, number 4, pages 1369–1379, October 2015. (**IEEE Best Paper Award**: [link](#))
- [4] **Jun Chen** and Ratnesh Kumar, “[Stochastic Failure Prognosability of Discrete Event Systems](#),” *IEEE Transactions on Automatic Control*, volume 60, number 6, pages 1570–1581, June 2015.
- [3] **Jun Chen** and Ratnesh Kumar, “[Failure Detection Framework for Stochastic Discrete Event Systems with Guaranteed Error Bounds](#),” *IEEE Transactions on Automatic Control*, volume 60, number 6, pages 1542–1553, June 2015.

- [2] **Jun Chen** and Ratnesh Kumar, “Polynomial Test for Stochastic Diagnosability of Discrete Event Systems,” *IEEE Transactions on Automation Science and Engineering*, volume 10, number 4, pages 969–979, October 2013.
- [1] Lingfei Zhi, **Jun Chen**, Peter Molnar and Aman Behal, “Weighted Least-Squares Approach for Identification of a Reduced-Order Adaptive Neuronal Model,” *IEEE Transactions on Neural Networks and Learning Systems*, volume 23, number 5, pages 834–840, May 2012.

#### Peer Reviewed Conference Articles

- [50] Ali Irshayyid and **Jun Chen\***, “Battery State-of-Health Estimation based on Partial Discharge Curves,” *IEEE Conference on Control Technology and Applications*, Vancouver, BC, Canada, August 12–16, 2026.
- [49] Luke Nukulaj, Wanqun Yang and **Jun Chen\***, “Robust Data-Enabled Predictive Control for EV Charging under Thermal Constraints,” *IEEE Transportation Electrification Conference & Expo*, Novi, MI, June 10–12, 2026.
- [48] Wanqun Yang, Luke Nukulaj and **Jun Chen\***, “Gain-Scheduled Data-Enabled Predictive Control for EV Thermal Management,” *IEEE Transportation Electrification Conference & Expo*, Novi, MI, June 10–12, 2026.
- [47] Ali Irshayyid, Wanqun Yang, Tingjun Lei and **Jun Chen\***, “Reinforcement Learning-Based Real-Time Balancing of Reconfigurable Battery Packs,” *IEEE Transportation Electrification Conference & Expo*, Novi, MI, June 10–12, 2026.
- [46] Briana Popa, Ali Irshayyid and **Jun Chen\***, “Lane Feature-Based Localization via ICP Map Alignment,” *North American International Conference on Industrial Engineering and Operations Management*, Milwaukee, Wisconsin, USA, June 9–11, 2026.
- [45] Luke Nukulaj and **Jun Chen\***, “Dense Extended Kalman Filter for Simultaneous Cell State-Parameter Estimation,” *American Control Conference*, New Orleans, LA, May 26–29, 2026. (**ATS Best Paper Finalist Award**: [link](#))
- [44] Zhaodong Zhou, Mingyuan Tao, Jiayi Qiu, Peng Zhang, Meng Xu, Yong Sun and **Jun Chen\***, “Personalized MPC for Autonomous Vehicle Lateral Motion Control via Inverse Reinforcement Learning,” *American Control Conference*, New Orleans, LA, May 26–29, 2026.
- [43] **Jun Chen\*** and Chong Li, “Asynchronous Model Predictive Control for Large Interconnected Systems with External Disturbance,” *American Control Conference*, New Orleans, LA, May 26–29, 2026.
- [42] Yunge Li, Zhaodong Zhou, Shaibal Saha, Ali Irshayyid, Keer Chen, Lanyu Xu and **Jun Chen**, “LightAD: A Lightweight Vision-Based Autonomous Driving System,” *IEEE International Conference on Mobility: Operations, Services, and Technologies*, Detroit, MI, May 4–6, 2026.
- [41] Artem Abzaliev, Rutchanon Hatasen, Hussein Kokash, Linda Zhu, **Jun Chen** and Mihai G. Burzo, “Machine Learning and Sensory Integration for Real-Time Road Surface Assessment,” *ASME International Mechanical Engineering Congress & Exposition*, Memphis, TN, USA, November 16–20, 2025.
- [40] Ashwin Devanga, Tingjun Lei\*, Jueming Hu, **Jun Chen** and Chaomin Luo, “Trustworthy Cyber-Resilient Reinforcement Learning for Secure Navigation under Adversarial Attack,” *IEEE Cyber Awareness and Research Symposium*, Grand Forks, ND, October 27–30, 2025. (**Best Paper Award**: [link](#))
- [39] Wanqun Yang, Mohammad R. Hajidavalloo, Zhaojian Li and **Jun Chen\***, “NMPC-based Cell-Level Thermal Management of EV Batteries in Low Temperature Environment,” *Modeling, Estimation and Control Conference*, Pittsburgh, PA, USA, October 5–8, 2025.
- [38] Hussein Alawsi, Zhaodong Zhou, Ali Irshayyid and **Jun Chen\***, “RL-assisted Model Predictive Control for Automated Parking Systems,” *IEEE International Conference on Unmanned Systems*, Changzhou, China, September 18–19, 2025.
- [37] Ali Irshayyid and **Jun Chen\***, “GNN-Based Surrogate Model for Reconfigurable Battery Packs,” *IEEE Conference on Control Technology and Applications*, San Diego, CA, August 25–27, 2025.
- [36] Lateefa Shibah Tusuubira, Wen-Chiao Lin\* and **Jun Chen**, “Fault Mitigation for Autonomous Vehicles with Reduced Front Steering Capability,” *IEEE International Conference on Prognostics and Health Management*, Denver, CO, June 9–11, 2025, 2025.

- [35] Cory Ness and **Jun Chen\***, “[Extended Kalman Filter for Flywheel Systems](#),” *IEEE International Conference on Electro Information Technology*, Valparaiso, IN, May 29–31, 2025.
- [34] Zhaodong Zhou, Yu Jiang and **Jun Chen\***, “[Exploring Optimal Pumping Strategy in Active Suspension Systems for Speed Maximization during Downhill Motion](#),” *12th IFAC Symposium on Intelligent Autonomous Vehicles*, Phoenix, AZ, May 7–9, 2025.
- [33] Luke Nculaj, Adam Kidwell, Connor Homayouni, Alex Fillmore, Darrin Hanna and **Jun Chen**, “[Optimal FPGA Implementation of Dense Extended Kalman Filter for Simultaneous Cell State Estimation](#),” *IEEE International Midwest Symposium on Circuits and Systems (MWSCAS)*, Springfield, MA, August 11–14, 2024.
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